Breath Check setup.

What you need to make it work.

Before hands you need a Windows 11 Laptop with a microphone,
Administrator Credentials, Internet access, libraries and frameworks.

Connect to windows as Administrator and work always as administrator.

Libraries and frameworks to install.

Please create a directory C:\Framework, change the properties and allow full control for all users.

In the C:\Framework directory create 3 sub directories:

  \FFMPEG
  \Python
  \nodejs
  \ngrok

Download the correct windows installer 64 bit .msi from:

👉 https://www.gyan.dev/ffmpeg/builds/

Download:

👉 “ffmpeg-release-full.7z” (or latest full build)

https://www.python.org/downloads/
https://nodejs.org/en
https://ngrok.com/download/windows

NGROK is required only if you want the software to work from a mobile.
As this is still a work in progress you can install it now, but it will not be used until the next update.

The prototype of software will work from a browser in your local pc.

Install all of the above in the respective directories you created.

For FFMPEG copy the \BIN directory and the rest in your FFMPEG.

in Windows search on the bottom bar type: Edit environment variables for your account

Press environmental variables.

Now you see the environments variables.

Select PATH then button Edit.

Press New and add one by one the path to your frameworks directories:

C:\Frameworks\ffmpeg\bin
C:\Frameworks\Python\bin
C:\Frameworks\Python\Scripts
C:\Frameworks\ngrok

Press ok and exit. Restart your laptop. Reconnect as administrator.

on disk c: do as you did before and create another folder

```
mkdir breath-check
cd breath-check

mkdir backend
mkdir frontend
```

now change the properties to the breath-check to allow full control to all users.

Open a command prompt, type cmd in the serch bar and run as administrator.

Go to backend folder

```
c:

CD c:\breath-check\backend
```

create virtual environment, type as follows:

```
python -m venv venv
```

press "enter"

```
venv\Scripts\activate
```

press "enter"

Leave the window open but minimized on the task bar.

open another command prompt as administrator. Go to backend directory.

```
c:

CD c:\breath-check\backend
```

Install dependencies, type as follows and press enter at the end:

```
pip install fastapi uvicorn librosa numpy scipy soundfile python-multipart pydub
```

Check there is no errors.

Open Notepad and create in backend the file main.py, copy and paste everything between asterisk, save the file.

****

```
from fastapi.middleware.cors import CORSMiddleware
from fastapi import FastAPI, File, UploadFile
import numpy as np
import librosa
from pydub import AudioSegment
import tempfile
import os

app = FastAPI()
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],  # allow all (OK for hackathon)
    allow_credentials=True,
```

```
    allow_methods=["*"],
    allow_headers=["*"],
)

# ----------------------------
# Simple AI-like analysis
# ----------------------------
def analyze_audio(y, sr):
    energy = np.mean(y**2)
    zcr = np.mean(librosa.feature.zero_crossing_rate(y))
    spectral_centroid = np.mean(librosa.feature.spectral_centroid(y=y, sr=sr))

    print(f"DEBUG -> Energy: {energy}, ZCR: {zcr}, Spectral: {spectral_centroid}")

    if energy < 0.001:
        return "⚠ Low breath strength detected"

    if spectral_centroid > 3000:
        return "⚠ Possible wheeze detected"

    if zcr > 0.15:
        return "⚠ Irregular breathing pattern"

    return "☑ Normal breathing pattern"


# ----------------------------
# API endpoint
# ----------------------------
@app.post("/analyze")
async def analyze(file: UploadFile = File(...)):
    try:
        # Read uploaded audio
        audio_bytes = await file.read()

        # Save as .webm (browser format)
        with tempfile.NamedTemporaryFile(delete=False, suffix=".webm") as tmp:
            tmp.write(audio_bytes)
            webm_path = tmp.name

        print(f"Saved WEBM: {webm_path}")

        # Convert to WAV using ffmpeg (via pydub)
        wav_path = webm_path.replace(".webm", ".wav")

        audio = AudioSegment.from_file(webm_path, format="webm")
        audio.export(wav_path, format="wav")

        print(f"Converted WAV: {wav_path}")

        # Load audio
        y, sr = librosa.load(wav_path, sr=None)

        print(f"Loaded audio: {len(y)} samples, SR: {sr}")

        # Run analysis
        result = analyze_audio(y, sr)

        # Clean up temp files
        os.remove(webm_path)
```

```
        os.remove(wav_path)

        return {"result": result}

    except Exception as e:
        print("ERROR:", str(e))
        return {"result": "❌ Error processing audio"}
```

****

in backend run the backend software

```
uvicorn main:app --reload
```

In a browser open:

http://127.0.0.1:8000/docs

You now have a working API.

Leave the window open but minimized on the task bar.

open another command prompt as administrator. Go to backend directory.

```
c:

CD c:\breath-check\frontend
```

Open Notepad and create in frontend the file index.html, copy and paste everything between asterisk, save the file.

****

```
<!DOCTYPE html>
<html>
<head>
  <title>Breath Check</title>
</head>
<body>

<h1>2-Minute Breath Check</h1>

<button onclick="startRecording()">Start Recording</button>
<button onclick="stopRecording()">Stop</button>

<p id="status">Status: idle</p>
<p id="result"></p>

<script>
let mediaRecorder;
let audioChunks = [];

async function startRecording() {
  document.getElementById("status").innerText = "Recording...";

  const stream = await navigator.mediaDevices.getUserMedia({ audio: true });
  mediaRecorder = new MediaRecorder(stream);

  audioChunks = [];
```

```
  mediaRecorder.ondataavailable = event => {
    console.log("Chunk received");
    audioChunks.push(event.data);
  };

  mediaRecorder.start();
}

async function stopRecording() {
  if (!mediaRecorder) {
    alert("Recording not started!");
    return;
  }

  document.getElementById("status").innerText = "Processing...";

  mediaRecorder.stop();

  mediaRecorder.onstop = async () => {
    const audioBlob = new Blob(audioChunks, { type: 'audio/webm' });

    const formData = new FormData();
    formData.append("file", audioBlob);

    try {
      const response = await fetch("http://127.0.0.1:8000/analyze", {
        method: "POST",
        body: formData
      });

      const data = await response.json();
      document.getElementById("result").innerText = data.result;

      document.getElementById("status").innerText = "Done";
    } catch (error) {
      console.error(error);
      document.getElementById("status").innerText = "Error";
    }
  };
}
</script>

</body>
</html>
```

****

Open Windows Defender Firewall

Chose Allow and App or feature through Windows Defender Firewall

Chose python and allow everything

Optionally you should find also ports 8000 and 8080 you need to allow for both

Confirm you allowed traffic with ok

in the command window of the frontend run

```
cd c:\breathcheck\frontend
```

python -m http.server 8080

open a browser and type in the address bar http://localhost:8080

🧪 ☑ SIMPLE TEST PROTOCOL (what YOU should do)

You want a structured 3-step test (like a real screening tool)

🧪 STEP 1 — Breathing test (10 seconds)
What to do:
Hold phone ~20 cm from mouth
Quiet room
Do:

👉 3 deep breaths:

INHALE (3 sec) → EXHALE (4 sec)

Repeat 3 times (~10 seconds total)

What the app is “looking for”:
airflow consistency
noise (wheezing-like sounds)
breath interruptions
Expected result:
Normal → low energy variation
No spikes → “green / normal”
🦠 STEP 2 — Cough test (5–8 seconds)
What to do:

👉 Do 3 strong coughs, like:

COUGH — pause — COUGH — pause — COUGH
Important:
Not fake throat clearing ❌
Real chest cough ☑
What the app detects:
sharp spikes in amplitude
spectral changes (cough signature)
Expected:
Strong spikes → classified as “cough detected”
🗣 STEP 3 — Voice test (5–10 seconds)
Say clearly:
"Today I feel well and my breathing is normal"
Why this matters:
voice stability
breath control while speaking
pauses / strain
🧠 What your prototype is ACTUALLY doing

Right now your backend is likely using:

amplitude (loudness)
zero-crossing rate
maybe spectral features

👉 So you're detecting patterns, not diagnosing

✐ FULL TEST FLOW (what to actually do)
Open app on phone
Press Start Recording
Do:

👉 10 sec breathing
👉 3 coughs
👉 say the sentence

Press Stop
Wait for result